

Report Proposal: IT Solution for Grocery Retail Online Business

By Shong QIEN BUO, Irwin Tan, and Calvine

SHONG QIEN BUO : M44100768

IRWIN TAN : M44100798

CALVINE : M44100604

Introduction:

Company Background

PV MART is a medium-sized grocery company in Malaysia. It was founded in 2007 and established in the Malaysian state of Selangor. Travis Porter, who is the founder of PV MART, and without his wife's support, will not be a success for this mart. In addition, it is extremely popular among the community and extremely competitive compared to its own business field, for example, like Tesco and Jaya Grocer. PV Mart sells essential daily items like milk, fresh vegetables, meat, all types of protein foods, kids' toys, gardening tools, clothes, stationery, etc.



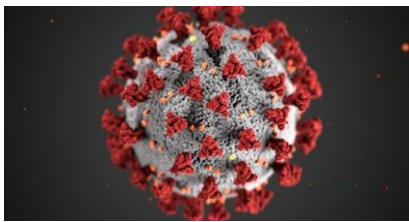
JAYA GROCER

TESCO



Problem Statement

In 2020, the COVID-19 pandemic struck a lot of grocery retail companies, including PV Mart. Due to the lockdown, customers were restricted from traveling, which brought a major impact on physical stores like PV Mart. For example, there is no online platform to deliver their products to the consumer, which stalls sales, equivalent in less revenue received and facing challenges in PV Mart's operation. Because of those problems, PV Mart made its main objective to convert their operations system to an online retail market.



COVID-19



Objectives of the proposal

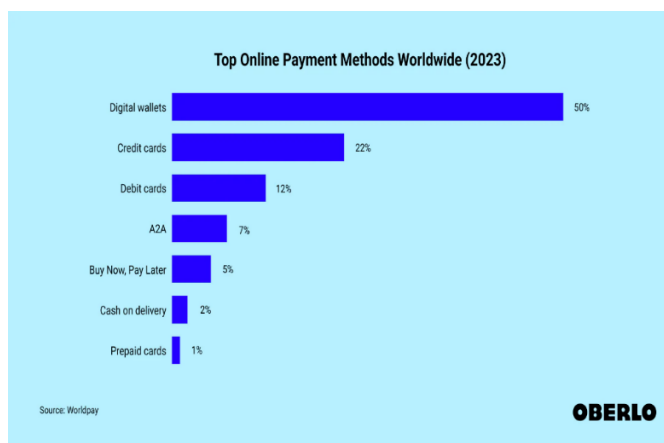
As an IT consultant, we can develop and launch an E-commerce platform using our IT skills to support customer search products and place online payments. This allows the process of the transaction to become faster and more efficient. Other than that, the user can also save their time while buying online, which delivers the products to the customer's doorstep. Furthermore, we can also avoid interaction as the government law states that people should maintain social distancing. By launching E-commerce, it also helps attract new customers who choose online platforms rather than physical shopping.



E-COMMERCE

Descriptions of computer-related operations

PV Mart's daily activities include collecting payments, creating shopping lists, selling products, handling deliveries, etc. Now, many of the tasks are still done manually or with basic digital applications such as Microsoft Excel and PDF files. This approach increases the chance of making mistakes and creates a less satisfactory experience for customers. To solve these issues and improve efficiency, Fresh Mart plans to upgrade its operations by introducing several IT-based systems. For example, PV MART creates a user-friendly website that displays all the selling products, online payment options, a search button, a digital shopping cart, and much more. In addition, this website provides a 24/7 service so consumers can shop anytime and anywhere without worrying about the business operations of PV MART. For online payment, the system can support many secure payment methods such as credit/debit cards, e-wallets, payWave, and online banking. This not only enhances convenience for customers but also reduces the amount of cash payment and saves time while conducting payment. For customer service, the website provides a 24/7 AI chatbot to help customers solve their problems while online shopping. For example, if customers want to know more about the latest information, items, or promotions on the website. Customers can just type their questions in the chatbot. Then, the chatbot will answer it accurately and fast. By using that, customers don't need to wait for staff availability, and it satisfies customers' expectations. Moreover, AI can also help customers to keep track of their shopping lists so that they won't forget to buy some items after checkout.



Top Online Payment Methods Worldwide (2023)



AI CHATBOT

Software requirements:

While all other online markets, PV market must set up some of their software to provide safety and improve efficiency. The most important requirement for all is to provide user verification, which is essential to avoid scams. The other factor that affects users is investigating user experience, which features must be clearly designed and easy to use without technical issues. Moreover, systems must specify how the payments go through accurately to avoid accidentally paying extra, double payments, or payment errors. Lastly, the PV market must be linked to an email or other social media on the website, just in case their users want to contact customer support, request refunds, or report financial problems. With all the software features, this increases trustworthiness for the online payment experience. By applying all these beneficial software requirements, PV Mart can build a reliable and secure online store that offers a smooth and satisfying experience for all its customers.



Hardware Requirements

Computers are essential for running software such as POS SYSTEMS (POINT OF SALE), accounting programs, and inventory management applications. Staff always use them to process sales, record data, and communicate with suppliers or customers. For example, staff use a reliable computer (like an all-in-one touchscreen POS system) with a sufficient processor and RAM, a solid-state drive (SSD) for speed and efficiency, a barcode scanner, and peripherals such as a cash drawer and a receipt printer to ensure smooth and secure payment. For network hardware requirements, PV MART must install a WIFI router that provides a high-speed connection across the entire store. In addition, PV MART will install a firewall or antivirus protection to secure some important files, data and customer's privacy data from cyberthreats. The market also has a centralized server system to store company files, customers' data, and website data securely.



Social and economic impact

The digital transformation of PV Mart through online platform development creates multiple positive effects on both social and economic impact. The system enables customers to purchase products from home without physical interaction, which protects them from transmission of COVID-19 while following social distancing guidelines. The system enables people with active schedules and elderly citizens and family groups to shop for groceries without needing to leave their homes. The online system enables PV Mart to expand its customer base beyond its physical store boundaries, which results in higher profits and lower operational expenses. The online system generates employment opportunities in IT and logistics and customer service who people with active schedules and elderly citizens, and family groups to shop for groceries without needing to leave their homes. The online system enables PV Mart to expand its customer base beyond its physical store boundaries which results in higher profits and lower operational expenses. The online system generates employment opportunities in IT and logistics and customer service which stimulates economic growth in the local area. The e-commerce platform deployment at PV Mart enables business growth while fostering technological advancement in society through enhanced connectivity and operational efficiency.



SOCIAL AND ECONOMIC IMPACT

Social and economic impact

The digital transformation of PV Mart through online platform development creates multiple positive effects on both social and economic impact. The system enables customers to purchase products from home without physical interaction, which protects them from transmission of COVID-19 while following social distancing guidelines. The system enables people with active schedules and elderly citizens and family groups to shop for groceries without needing to leave their homes. The online system enables PV Mart to expand its customer base beyond its physical store boundaries, which results in higher profits and lower operational expenses. The online system generates employment opportunities in IT and logistics and customer service who people with active schedules and elderly citizens, and family groups to shop for groceries without needing to leave their homes. The online system enables PV Mart to expand its customer base beyond its physical store boundaries which results in higher profits and lower operational expenses. The online system generates employment opportunities in IT and logistics and customer service which stimulates economic growth in the local area. The e-commerce platform deployment at PV Mart enables business growth while fostering technological advancement in society through enhanced connectivity and operational efficiency.



SOCIAL AND ECONOMIC IMPACT

